

to topic: Expectation - Maximization (EM) algorithm

gaussian distribution (normal distribution)

A gaussian distribution is a continuous probability distribution that describes how values are distributed around a mean.

It is also called the normal distribution.

It has a bell-shaped curve.

→ Mathematical formula

$$N(x | \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

where:

- μ = mean (centre)
- σ^2 = Variance (spread) → box
- σ = standard deviation

large variance → wide curve

small variance → narrow curve

symmetric around mean

total area under curve = 1

mean = median = mode

gaussian mixture Model (GMM)

sometimes data does not come from one gaussian.

instead:

it comes from multiple gaussians combined.

$$p(x) = \sum_{k=1}^K \pi_k N(x | \mu_k, \sigma_k^2)$$

where :

k = no. of clusters

π_k = mixing coefficient

$$\sum \pi_k = 1$$

This is called GMM.

Q. why do we need EM algorithm?

In GMM:

we do not know :

- which data point belongs to which gaussian
- true means • true variance
- true mixing coefficient

so, we treat cluster membership as a hidden variable
EM helps estimate these unknown parameters.

EM algorithm

EM is an iterative optimization algo and when there are hidden variables.

It maximizes the likelihood of data.

→ EM has 2 steps:

step 1:- E-steps (Expectation step)

we calculate probability that each point belongs to each cluster.

This is called responsibility?

$$r_{ik} = \frac{\pi_k N(x_i | \mu_k, \sigma_k^2)}{\sum_j \pi_j N(x_i | \mu_j, \sigma_j^2)}$$

It tells :

probability that point x_i belongs to cluster k .

Step 2:

M-step (Maximization step)

now update parameters using responsibilities

effective no. of points in cluster k :

$$N_k = \sum_i \gamma_{ik}$$

update mean:

$$\mu_k = \frac{1}{N_k} \sum_i \gamma_{ik} x_i$$

update variance:

$$\sigma_k^2 = \frac{1}{N_k} \sum_i \gamma_{ik} (x_i - \mu_k)^2$$

update mixing coefficient:

$$\pi_k = \frac{N_k}{n}$$

then repeat E-step & M step until convergence.

How EM works intuitively

1. guess cluster centers
2. Assign soft probability
3. Recalculate cluster centers
4. Repeat
5. Eventually parameters stop changing.

why EM converges?

- each iteration increase log-likelihood
- it moves toward local maximum
- stop when parameter stabilize

Q1. 1D dataset $\Rightarrow X = \{1, 2, 8, 9\}$, $K=2$

mixture coefficients: $\pi_1 = 0.5$, $\pi_2 = 0.5$, Mean: $\mu_1 = 2$, $\mu_2 = 8$.

Variances $\Rightarrow \sigma_1^2 = 1$, $\sigma_2^2 = 1$

E step (iteration-1)

for $\pi = 1$

$$N(1/2, 1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(1-2)^2}{2}} = \frac{1}{\sqrt{2\pi}} e^{-1/2}$$

for cluster 2:

$$N(1/8, 1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(1-8)^2}{2}} = \frac{1}{\sqrt{2\pi}} e^{-49/2}$$

very small value

$$\text{so, } \gamma_{11} = \frac{0.5 \times \text{large}}{0.5 \times \text{large} + 0.5 \times \text{tiny}} \approx 1$$

$$\gamma_{12} \approx 0$$

similarly :

$x=2 \rightarrow \text{cluster 1}$

$x=8 \rightarrow \text{cluster 2}$

$x=9 \rightarrow \text{cluster 2}$

so responsibility become :

X	y_1	y_2
1	1	0
2	1	0
8	0	1
9	0	1

M-step

$N_1 = 2$, $N_2 = 2$

update means

$$\mu_1 = \frac{1 \times 1 + 2 \times 1 + 8 \times 0 + 9 \times 0}{2} = 1.5$$

$$\mu_2 = \frac{8 + 9}{2} = 8.5$$

update variance

$$\sigma_1^2 = \frac{(1-1.5)^2 + (2-1.5)^2}{2} = \frac{0.25 + 0.25}{2} = 0.25$$

$$\sigma_2^2 = 0.25$$

update mixing coefficient

$$\pi_1 = \frac{2}{4} = 0.5$$

$$\pi_2 = 0.5$$

now, means are 1.5 & 8.5

- gaussian operation becomes even stronger
- responsibilities stay same
- so, parameter no longer change.
- converged after iteration 2.

question: Dataset : {2, 3, 4, 10, 11, 12}

initial : $\pi_1 = \pi_2 = 0.5$
 $\mu_1 = 3, \mu_2 = 11$
 $\sigma_1^2 = \sigma_2^2 = 1$

E step (iteration 1)
 compute responsibilities.

for $x=2$:

$$N(2/3, 1) = \frac{1}{\sqrt{2\pi}} e^{-1/2}$$

$$N(2/11, 1) = \frac{1}{\sqrt{2\pi}} e^{-81/2}$$

$$y_{21} \approx 1$$

similarly: 3, 3, 4 $\rightarrow$ cluster 1, 10, 11, 12 $\rightarrow$ cluster 2

responsibilities:

x	y_1	y_2
2	1	0
3	1	0
4	1	0
10	0	1
11	0	1
12	0	1

M-step $N_1 = 3, N_2 = 3$ update means $\mu_1 = 3$
 $\mu_2 = 11$

update variance:

$$\sigma_1^2 = \frac{(2-3)^2 + (3-3)^2 + (4-3)^2}{3} = 0.6667$$

$$\sigma_1^2 = 0.6667$$

mixing coefficient $\pi_1 = 0.5$, $\pi_2 = 0.5$

#

code

```
double gaussian(double x, double mean, double variance){
    double coeff = 1.0 / sqrt(2 * M_PI * variance);
    double exponen = exp(-(x - mean) * (x - mean) / (2 * var));
    return coeff * exponen;
}
```

• for E-step:-

```
for (int i = 0; i < n; i++) {
```

```
    double denom = 0;
```

```
    for (int k = 0; k < K; k++)
```

```
    {
```

```
        gamma[i][k] = pi[k] * gaussian(x[i], mu[k], var[k]);
```

```
        denom += gamma[i][k];
```

```
    }
```

```
    for (int k = 0; k < K; k++)
```

```
        gamma[i][k] /= denom;
```

```
    }
```

```
}
```

• M-step:-

```
for (int k = 0; k < K; k++)
```

```
{
```

```
    double nk = 0;
```

```
    double N_mean_k = 0;
```



```
for (int i=0; i < n; i++)
```

```
    Nk += gamma[i][k];
```

```
    newMean += gamma[i][k] * x[i];
```

```
}
```

```
newMean /= Nk;
```

```
for (int i=0; i < n; i++)
```

```
    newVar += gamma[i][k] * pow(x[i] - newMean, 2);
```

```
newVar /= Nk;
```

• convergence check

```
if (fabs(mu[k] - old_mu[k]) > epsilon ||
```

```
    fabs(var[k] - old_var[k]) > epsilon ||
```

```
    fabs(pi[k] - old_pi[k]) > epsilon)
```

```
    converged = false;
```

```
    break;
```

```
}
```

